

Steven Tin Sui Luo – Curriculum Vitae

CONTACT INFORMATION	38 Iannuzzi St Toronto, ON M5V 0S2, Canada	<i>Tel:</i> (647) 674-1053 <i>Email:</i> stevents.luo@mail.utoronto.ca <i>Website:</i> https://stevolopolis.github.io
RESEARCH INTERESTS	Areas: Machine learning, brain & cognitively inspired learning, test-time learning, machine memory, machine lifelong learning, neural fields, computer vision My key research question is how can we enable machines to autonomously adapt in the wild without human intervention? Humans could derive their own goals, acquire their own data, and seamlessly switch between different timescales of learning and memory formation. I'm particularly interested in developing a unified connectionist architecture that could internalize these components of intelligence to enable continuous adaptation.	
EDUCATION	University of Toronto (St. George), Canada <i>Department of Computer Science</i> 2021/09 – Present Computer Science Specialist & Math Major	
	St Paul's Co-educational College, Hong Kong International Baccalaureate (IB): 45/45 2015/09 – 2021/06	
PROFESSIONAL EXPERIENCE	Vivid Machines , Toronto, ON, Canada 2024/05 – 2024/12 <i>Deep Learning Associate, Supervisor: Pegah Kamousi</i> Improved robustness and generalization of tree detection model Iterated 3 YOLO models to production and reduced tree detection error from 50% to 15%	
	EN:ai , Hong Kong, China 2021/05 – 2021/08 <i>Machine Learning Intern, Supervisor: Ngai Wong</i> Developed hand detection and hand-keypoint detection model using single-shot detector and mobileNetV2 Achieved real-time inferencing (20+ fps) on cpu	
RESEARCH EXPERIENCE	Agentic Learning AI Lab , NYU 2025/05 – Present <i>Visiting Student Researcher, Supervisor: Mengye Ren</i> Investigating the use of gisting and linear RNNs to enable in-context concept learning and infinite-shot in-context learning in language models.	
	Toronto Computational Imaging Group , UofT 2024/01 – 2024/05 <i>Undergraduate Researcher, Supervisor: David Lindell</i> Developed a theory of domain manipulation that explained the effectiveness of grid-based neural fields. Technical report received A+ for CSC494 Thesis Course.	
	NGai Lab , HKU 2023/05 – 2024/04 <i>Visiting Student Researcher, Supervisor: Ngai Wong</i> Developed an architecture for neural fields that decouples inference cost with network depth. Realized the nonparametric teaching theory on neural fields to speed up training time by up to 1.5×. Two papers accepted at ICLR24 and ICML24 , respectively.	
	Cognitive Neuroscience and Sensorimotor Integration Lab , UofT Scarborough 2021/09 – 2022/12 <i>Undergraduate Researcher, Supervisor: Matthias Niemeier</i> Developed a novel loss function and an architecture to unify grasping and detection CNNs under identical training regime, enabling pairwise comparison with the EEG signals from human brains Preliminary poster accepted to MAIN 2022 Final manuscript to be submitted to Nature Neuroscience .	

PEER-REVIEWED CONFERENCE PUBLICATIONS	(*=equal contribution)	
	Zhang, Chen, Steven Tin Sui Luo* , Jason Chun Lok Li, Yik-Chung Wu, and Ngai Wong. "Non-parametric teaching of implicit neural representations." <i>In Proceedings of the 41st International Conference on Machine Learning</i> , 2024.	
	Li, Jason Chun Lok, Steven Tin Sui Luo* , Le Xu, and Ngai Wong. "ASMR: Activation-Sharing Multi-Resolution Coordinate Networks for Efficient Inference." <i>In The Twelfth International Conference on Learning Representations</i> , 2024	
PREPRINTS & REPORTS	Reza, Tahsin, Ewan Jordan, Steven Tin Sui Luo , Kirtan Patel, Jessica Tang, Matthias Niemeier. "From Tasks to Topology: Dorsal and Ventral Streams Emerge in Optimized Neural Networks." <i>preprint</i> , 2025	
	Steven Tin Sui Luo . "A New Perspective To Understanding Multi-resolution Hash Encoding For Neural Fields." <i>arXiv preprint arXiv:2405.10531</i> , 2025	
	Tahsin, Reza, Steven Tin Sui Luo* , Rohan Jain, Jack Cai, and Matthias Niemeier. "Task-Agnostic Approach to Modeling the Ventral and Dorsal Stream." <i>In The 6th Edition Of The Montreal AI And Neuroscience Conference</i> , 2022	
HONORS AND AWARDS	Faulds Janet Elizabeth Admissions Award (\$2000)	2021
	St. Michael's College Travel Grant (\$750)	2024
	Department of Computer Science Academic Travel Grant (\$600)	2024
	ASSU Travel Grant (\$300)	2024
	2-time Dean's List Scholar	2021-2025
SERVICE	Founder/Organizer: EigenAI Student Conference	2023
	VP of Engineering: University of Toronto Machine Intelligence Student Team	2022
REFERENCES	Dr. Mengye Ren , Assistant Professor of Computer Science and Data Science, New York University, +1 (212) 992-7547, mengye@cs.nyu.edu	
	Dr. Ngai Wong , Associate Professor of Electrical and Electronic Engineering, The University of Hong Kong, +852 3917-1914, nwong@eee.hku.hk	
	Dr. David Lindell , Assistant Professor of Computer Science, University of Toronto, +1 (507) 514-2491, lindell@cs.toronto.edu	